

Statement of Work
Rack-Mountable, High-Power, Frequency-Doubled Red Laser System for Calcium
Clock Development Division, Precise Time Department
U. S. Naval Observatory

BACKGROUND

The United States Naval Observatory (USNO) is the authoritative source of time for the Department of Defense. For this purpose, the USNO's Precise Time Department maintains an ensemble of high-performing atomic clocks that is used to generate precise time. Because of the lack of commercial vendors for the highest performing clocks, stocking the clock ensemble with systems based on cutting-edge technology requires that these systems be designed and built in-house.

Within the Precise Time Department, the Clock Development Division is investigating a transition within calcium that can be driven with 657nm laser light serves as the frequency reference. A standard external cavity diode laser (ECDL) at 657nm does not output enough power, and amplifiers at this wavelength are inadequate. To meet the optical power needs for our application, the laser system must generate 657nm via second harmonic generation (SHG). At the fundamental, a 1315nm ECDL amplified with a Tapered Amplifier (TA) is required to obtain a high 657nm output power from the doubling cavity. Additionally, high-bandwidth laser locking electronics are required to narrow the laser linewidth below the free-running linewidth of the laser system to hertz-level linewidths.

SCOPE

The USNO is seeking one high power 657nm laser system and one digital laser locking device. The full list of requirements is listed in the next section.

REQUIREMENTS

1. System needs to use an amplified 1315nm source laser that frequency doubles to a wavelength centered around 657.459nm
2. Linewidth: 50kHz or less
3. Mode-hop free tuning range about required wavelength: 15GHz
4. Generated 657nm output must be coupled into fiber with at least 600mW exiting the optical fiber.
5. The 1315nm light must be generated by an ECDL and TA configuration
6. Optical isolation: System must include optical isolation between the 1315nm source and subsequent amplifier and between the amplifier and the frequency doubler.
7. System must include a pick-off to be able to measure the wavelength of the source laser at 1315nm. This light must be fiber coupled and at a power of at least 3 mW.
8. System must include auto-tuning of the alignment of the source laser into the tapered amplifier and the amplified output into the SHG stage.
9. System must include auto-relocking of the resonant cavity used for frequency doubling, preferably using the Pound-Drever-Hall method of stabilization.

10. The system must include an electro-optic modulator for the SHG stage to remove residual frequency or amplitude modulation
11. The final spatial mode out of the frequency doubler must be well shaped and collimated.
12. The system must be 'turn-key' and able to achieve these specifications in a standard laboratory environment without user-supplied parts of assembly.
13. System must be rack-mountable in a standard 19" equipment rack and a fluid chiller included to temperature stabilize the system

Additionally, the vendor needs to provide a laser locking system that seamlessly interfaces with the laser controller and meets the following requirements:

14. Capable of locking to high finesse cavities or frequency combs and phase locking between lasers.
15. Bandwidth: 50 MHz
16. Linewidth: <1 Hz
17. PID regulator with 10ns or less signal delay

FACILITIES, EQUIPMENT, AND TRAVEL

The planning and support for the completion of these tasks will be coordinated with the technical POC and performed at USNO DC.

PERIOD OF PERFORMANCE

The period of performance is 5 months from the date of award.